AWS DEPLOYMENT & SUBSCRIPTION ECONOMICS

A practical path to launch and price FrameBucket

Planning model for the current photographer user module • India (Mumbai region) • 6 November 2026

RECOMMENDED START	REVENUE MODEL	PRICING DIRECTION
S3 + CloudFront, EC2 app, RDS PostgreSQL, one ALB. Start with GitHub Actions; defer EKS until it has an operational reason.	At the modeled 50/25/20/5 plan mix, current monthly ARPU is ₹859. The suggested menu averages ₹1,074.	Keep entry plans close to local alternatives; price the 500 GB tier for its storage cost and meter downloads/overages.

Suggested monthly package prices

Plan	Current / month	Included RAW storage	Suggested / month	Suggested annual total*
Bronze	₹599	None	**₹699**	₹7,549
Silver	₹799	50 GB	**₹999**	₹10,789
Gold	₹1,299	100 GB	**₹1,499**	₹16,189
Platinum	₹1,999	500 GB	**₹3,499**	₹37,789

*Annual total is 12 monthly payments less 10%, rounded to the nearest rupee. Prices are before GST. The recommendation assumes the listed product features and storage limits are delivered; it is a planning price, not a tested willingness-to-pay result.

Why these prices

A current India-focused photography studio product lists ₹599 for 20 GB, ₹1,199 for 50 GB and ₹1,499 for 100 GB. That is a useful reference, though feature sets differ. FrameBucket's planned 500 GB Platinum allowance needs a larger premium: at the modeled Mumbai S3 rate, a fully used 500 GB allowance costs about ₹1,180/month for storage alone. Meter unusually high downloads and offer storage add-ons after measuring real usage.

Market reference: Pixela published plan prices. S3 rate and storage sensitivity are detailed on page 4.

Important readiness note

The repository describes checkout as a preview: paid plan activation, payment processing and automatic trial expiry are not implemented. Until billing and entitlement handling are added, the modeled subscription revenue is forecast revenue, not cash currently collected.

01 / DEPLOYMENT DESIGN

Recommended AWS setup

Use the simplest production path that supports the current Spring Boot + React application. AWS cost figures below are monthly planning estimates, not a fixed quote.

GitHub Actions	Frontend	Backend	Database
Build on push to main. Authenticate to AWS with OIDC.	Vite build → private S3 bucket → CloudFront with HTTPS and your domain.	Docker image → EC2 instances in private subnets → ALB HTTPS endpoint.	RDS PostgreSQL in private subnets. Run Flyway migrations as a release step.

Sizing assumptions

Paid subscribers	EC2 app	EKS workers	EC2 database	RDS database
10	1 × t4g.medium	2 × t4g.medium	t4g.small + 30 GB	db.t4g.micro + 20 GB
100	2 × t4g.medium	2 × t4g.medium	t4g.medium + 100 GB	db.t4g.small + 50 GB
1,000	2 × t4g.large	3 × t4g.large	t4g.large + 200 GB	db.t4g.medium + 100 GB

For a small Spring Boot workload with modest concurrency and no photo processing. At 10 subscribers, one EC2 app instance is a low-cost starting point with no app-server failover. All database rows are Single-AZ. EKS workers are spread across two Availability Zones in the model.

Estimated monthly AWS infrastructure

Backend / database	10 subscribers	100 subscribers	1,000 subscribers
EC2 app + EC2 PostgreSQL	₹9,500	₹12,700	₹18,200
EC2 app + RDS PostgreSQL	₹10,100	₹13,650	₹20,200
EKS app + EC2 PostgreSQL	₹18,200	₹19,600	₹28,600
EKS app + RDS PostgreSQL	₹18,800	₹20,550	₹30,600

Shared layer budget: about ₹6,500/month for one ALB, one NAT Gateway, low-traffic static hosting/DNS and modest logs/backup. EKS totals include a standard-support cluster control plane; AWS bills the cluster separately from worker nodes. One NAT Gateway is the cost-optimized assumption; a second adds about ₹3,100/month. Traffic, storage, tax and payment fees are excluded.

02 / REVENUE & CONTRIBUTION

What the recommended path earns

The recommended path is EC2 for the app + RDS PostgreSQL. Revenue uses the assumed monthly mix: 50% Bronze, 25% Silver, 20% Gold and 5% Platinum. Contribution means revenue less estimated AWS infrastructure only.

Current menu: ₹599 / ₹799 / ₹1,299 / ₹1,999

Metric	10 subscribers	100 subscribers	1,000 subscribers
Revenue / month	₹8,590	₹85,900	₹8,59,000
AWS infrastructure	₹10,100	₹13,650	₹20,200
Contribution before photo storage	-₹1,510	₹72,250	₹8,38,800
Contribution if plan storage allowances are 100% full	-₹2,867	₹58,680	₹7,03,100

Suggested menu: ₹699 / ₹999 / ₹1,499 / ₹3,499

Metric	10 subscribers	100 subscribers	1,000 subscribers
Revenue / month	₹10,740	₹1,07,400	₹10,74,000
AWS infrastructure	₹10,100	₹13,650	₹20,200
Contribution before photo storage	₹640	₹93,750	₹10,53,800
Contribution if plan storage allowances are 100% full	-₹717	₹80,180	₹9,18,100

At the sample mix, suggested pricing raises average monthly revenue per subscriber from ₹859 to ₹1,074. Ten subscribers still sit near break-even once full storage use is allowed for; fixed load-balancer/network costs dominate at that scale. If all subscribers pay annually with the 10% discount, suggested-menu monthly-equivalent revenue is ₹9,666 / ₹96,660 / ₹9,66,600.

The 100%-full storage line is a conservative capacity scenario, not a usage forecast. It assumes each account uses all of its plan's quota. At 20% average quota use, the corresponding storage costs are about ₹270 / ₹2,710 / ₹27,140 per month for 10 / 100 / 1,000 accounts.

03 / DECISIONS & LIMITS

Protect the margin as the product grows

Photo storage sensitivity

Subscribers	Average max quota at modeled mix	S3 cost if full	S3 cost at 20% use
10	575 GB	₹1,360/mo	₹270/mo
100	5,750 GB	₹13,570/mo	₹2,710/mo
1,000	57,500 GB	₹1,35,700/mo	₹27,140/mo

Storage estimate uses the modeled plan mix and about ₹2.36 per GB-month for S3 Standard in Mumbai. It excludes requests, downloads and image delivery bandwidth; those can be material for photo products. The current user module does not yet implement photo upload/processing, so storage is excluded from the base infrastructure table.

Before go-live

1. Connect the purchased domain: private S3 origin behind CloudFront with Origin Access Control; HTTPS for the site. Use an API subdomain pointing to an ALB with HTTPS.

2. Use GitHub Actions with AWS OIDC. Build the Vite frontend, sync the bundle to S3 and invalidate CloudFront; build/push the backend image and deploy to EC2. Keep AWS keys out of repository secrets.

3. Keep RDS private and run Flyway once per release. Set the production frontend URL, SMTP/email settings, secure session cookies, CORS and CSRF allow-list.

4. Before running more than one backend instance, put HTTP sessions in a shared store (Spring Session JDBC with RDS or Redis). The current backend stores security context in HttpSession; stickiness alone does not preserve sessions if an instance fails.

5. Add payment-provider webhooks, paid-plan activation, renewal/cancellation handling and automatic trial expiry before treating subscriptions as revenue.

Risk and costing notes

The base estimate is Single-AZ for the database and uses one NAT Gateway. Add a second NAT Gateway for AZ-level egress resilience; use RDS Multi-AZ or a replicated PostgreSQL design when database availability warrants its additional cost. EKS is substantially more expensive at 10-100 users because the cluster fee is fixed. The estimate excludes payment processor fees, AWS taxes, labor, marketing, support and high data egress. Exact capacity depends on concurrent use and image-transfer patterns.

Pricing and product references

Project plan prices, 10% annual discount and unfinished checkout/trial behavior: fb_main/README.md and V4 subscription migration.
Market check: Pixela India pricing (published ₹599 / ₹1,199 / ₹1,499 tiers with 20 / 50 / 100 GB; features differ).
AWS pricing model: EKS • ALB • VPC / NAT / IPv4 • RDS PostgreSQL • S3.
Rate references: Mumbai EC2 rate list • Mumbai RDS rate list • USD/INR reference history. Rates are indicative and may not match discounts or the final AWS invoice.